

```
# Fallback: split by character limit if no separators work
return [text[i:i + max_chunk_size] for i in range(0,
len(text), max_chunk_size)]
```

Document-based chunking: Chunks are created from chunks based upon logical document boundaries (headings, subheadings, paragraphs, and sections), allowing the content's intended meaning as well as its structure to be retained, making this option the optimal method of chunking for any production RAG systems, particularly for technical, legal, or knowledge-based documents.

```
from typing import List
import re
def markdown_document_chunking(text: str) -> List[str]:
    # Split by markdown headers (# ## ### etc.)
    header_pattern = r'^#{1,6}\s+.+$'
    lines = text.split('\n')

    chunks = []
    current_chunk = []

    for line in lines:
        # Check if this line is a header
        if re.match(header_pattern, line, re.MULTILINE):
            # Save previous chunk if it has content
            if current_chunk:
                chunk_text = '\n'.join(current_chunk).strip()
                if chunk_text:
                    chunks.append(chunk_text)

            # Start new chunk with this header
            current_chunk = [line]
        else:
            # Add line to current chunk
            current_chunk.append(line)

    # Add final chunk
    if current_chunk:
        chunk_text = '\n'.join(current_chunk).strip()
        if chunk_text:
            chunks.append(chunk_text)

    return chunks
```

Embeddings and vector storage power semantic retrieval

Embeddings take content (such as text and audio data) and convert it into numeric arrays that represent the meaning of the data. Figure 4 illustrates that semantically related sentences create embeddings that are closer together than disparate sentences. Once you select your embedding model,

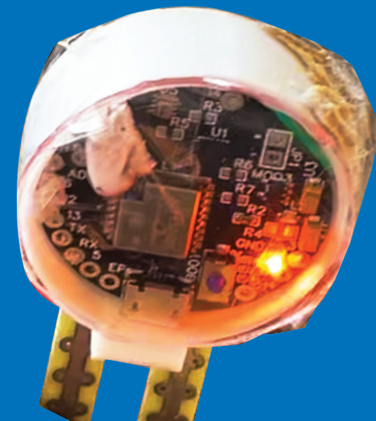
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